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My Research Program and Lab Focus on the Science of Engagement

WHAT IS ENGAGEMENT?

- Involves investment of energies:
  - *Affective:* moods, emotions, well-being
  - *Cognitive:* selective attention and task-related information processing
  - *Physical:* performing a task or activity
- Engagement with what?
  - Stimulus (e.g., text message prompt)
  - Task (e.g., go for a walk)

HOW CAN WE PROMOTE ENGAGEMENT?

- **Health behavior change can be enhanced**
  - Deliver intervention prompts in moments when engagement is likely
  - Personalize interventions based on internal states, external contexts, individual characteristics

ADAPTIVE TOOLS FOR ENAGAGEMENT

- **Just-in-time Adaptive Interventions (JITAIs) adapt to events that:**
  - change relatively rapidly (e.g., stress, location, alcohol use)
    - minutes, hours, days, weeks
  - represent risk for adverse outcomes (e.g., binge drinking) or opportunities to improve behaviors (e.g., self-monitor)
- **JITAIs provide** the right type of support, at the right time, in "the wild"
- **Unanswered questions about how to construct a JITAI,** can be addressed with experiments to "optimize" JITAIs
  - **Micro-randomized trial (MRT) designs**
    - Randomize individuals at points in time (e.g., every evening) when appropriate to deliver intervention (e.g., to self-monitor)
    - Determine when and for whom different intervention options are best

Promoting Engagement Through the Optimization of Digital Just-in-Time Adaptive Health Interventions

MRT FEASIBILITY STUDY TO DEVELOP ENGAGEMENT JITAI TO INCREASE WEAR TIME OF WRIST-WORN BIOSENSOR AMONG YOUNG ADULTS WITH HIV: LESSONS LEARNED

Young adults with HIV have disproportionately high rates of binge drinking and Alcohol Use Disorder

Wearable technologies aid in real-time self-monitoring

Skyn alcohol use biosensor is worn on wrist and measures transdermal alcohol concentration (TAC) based on alcohol from skin

Many people abandon wearable technologies quickly and after minimal use

Need to **identify strategies** to increase Skyn biosensor wear time:

- **Reciprocity** – small no-strings attached reward (e.g., $1; game points) activates human tendency to return favors and kind actions
- **Personalized feedback** – individualized feedback on Skyn biosensor wear time increases awareness of behavior and provides concrete point for improvement

Sample: 49 Young Adults (18-29) with HIV completed baseline session

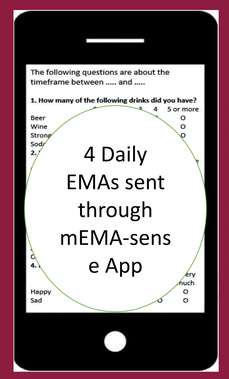
Condition (at onboarding): Standard (69%) or Engagement (31%)

Proximal outcome: Skyn biosensor wear time within last 12 hours

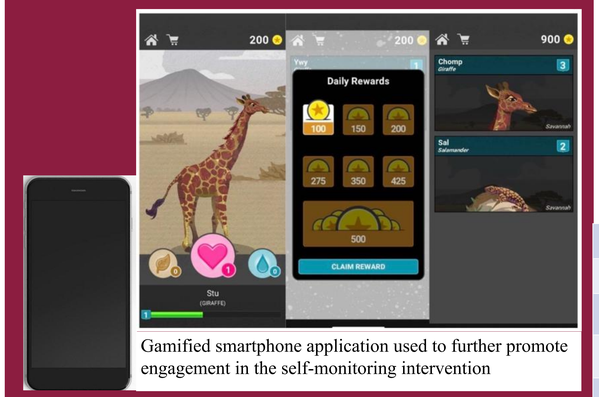
Distal outcome: Reduced binge drinking at 1 month assessment

Engagement Condition:

- Gamified Engagement Smartphone App Strategy
- Message Strategy

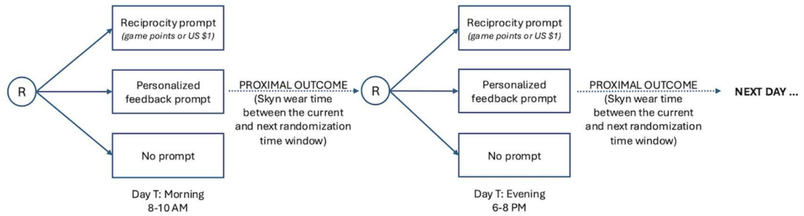


1 Morning & 3 Random ecological momentary assessments (EMAs) to collect real-time information on internal and external states



Gamified smartphone application used to further promote engagement in the self-monitoring intervention

Figure 2. Microrandomized trial design. Microrandomizations occur twice daily for 30 days.



Message Examples

Prompt Type	Message
Reciprocity (US $1)	Got the blues today? How about some green! (app)'s added a $1 gift to your Cash account.
Reciprocity (game points)	Hey there! Your animal(s) miss you. Here are 100 free bonus credits from (app) to your Ranch account. Treat them today!
Personalized Feedback	Look at you being productive! You wore your Skyn sensor for <<XX>> hours yesterday. Keep going!

PRELIMINARY RESULTS

Sample Characteristics	Mean (SD) or # (%)
Age	26.7 (± 2.5)
Race	
White	21 (42.9%)
Black	16 (32.6%)
Other	7 (14.3%)
Unknown/Did not answer	5 (10.2%)
Hispanic/Latinx	18 (36.7%)
Sex at birth	
Male	43 (87.8%)
Female	6 (12.2%)
Current sexual orientation	
Heterosexual	9 (18.4%)
Non-heterosexual	40 (81.6%)
Educational attainment	
High school grad or less	12 (24.5%)
Some college/Associates	27 (55.1%)
Bachelor's or higher	10 (20.4%)
Annual household income	
20k or less	14 (28.6%)
20,001 – 40k	10 (20.4%)
40,001 – 75k	6 (12.2%)
75,001 or more	2 (4.1%)
Unknown/Not collected	17 (34.7%)

20% attrition following baseline

Higher attrition (drop out) in standard condition:

- Standard condition: 37%
- Engagement condition: 17%

Reasons:

- 44% stopped responding
- 12% reported unstable housing (e.g., lived in a shelter, changed addresses mid-study) leading to lost or stolen equipment
- 16% reported a difficult financial situation or were concerned about the compensation schedule
- 28% difficulty with sensor and/or smartphone, or no reason given

Person-level factors associated with study retention

Variable	Non-Retained (Attrition)	Retained	Fisher's exact test
	Mean ± SD or %	Mean ± (SD) or %	p-value
Annual household income			
20k or less	71.4%	27.8%	
20,001 – 40k	0%	50.0%	
40,001 – 75k	28.6%	11.1%	
75,001 or more	0.0%	11.1%	0.035
Sensation seeking scale	4.2 ± 0.7	3.1 ± 1.0	0.024
Sexual sensation seeking scale	3.6 ± 0.3	3.3 ± 0.5	0.063†
DSM-5 mental health disorder symptoms	27.9 ± 14.1	15.9 ± 13.2	0.016
Cocaine use	4.3 ± 5.6	2.2 ± 6.0	0.063†
Pittsburgh sleep quality index (PSQI)	9.2 ± 3.3	7.0 ± 3.6	0.086†
Big-5 Extraversion	12.2 ± 1.9	9.2 ± 2.4	< 0.001

† Denotes marginal finding

CONCLUSIONS

- Retention lower without engagement strategies
- Person-level factors associated with retention
- Ongoing: MRT and usability analyses still underway. Stay tuned for results!

Lessons Learned Paper (in preparation):
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SCAN to Connect and Learn More About My Research